2024 Semester Two (October 2024) Examination Period

Faculty of Science

EXAM CODES:	**MAT9004**
TITLE OF PAPER:	Mathematical foundations for data science and AI
EXAM DURATION:	3 hours 10 mins

Rules

By starting your eExam, you're confirming that you're well enough to do the exam. If you become unwell during the assessment you must alert your online supervisor or contact the Exam Support hotline.

During your eExam, you must not have in your possession any item/material that has not been authorised for your exam. This includes books, notes, paper, electronic device/s, smart watch/device, or writing on any part of your body. Authorised items are listed above. Items/materials on your device, desk, chair, in your clothing or otherwise on your person will be deemed to be in your possession. Mobile phones must be switched off and placed face-down on your desk during your exam attempt.

You must not retain, copy, memorise or note down any exam content for personal use or to share with any other person by any means during or following your exam. You are not allowed to copy/paste text to or from external sources unless this has been authorised by your Chief Examiner.

You must comply with any instructions given to you by Monash exam staff.

As a student, and under Monash University's Student Academic Integrity procedure, you must undertake all your assessments with honesty and integrity. You must not allow anyone else to do work for you and you must not do any work for others. You must not contact, or attempt to contact, another person in an attempt to gain unfair advantage during your assessment. Assessors may take reasonable steps to check that your work displays the expected standards of academic integrity.

Failure to comply with the above instructions, or attempting to cheat or cheating in an assessment may constitute a breach of instructions under regulation 23 of the Monash University (Academic Board) Regulations or may constitute an act of academic misconduct under Part 7 of the Monash University (Council) Regulations.

<u>Authorised Materials</u>

CALCULATORS	☐ YES	☑ NO
DICTIONARIES	☐ YES	☑ NO
NOTES	☐ YES	☑ NO
WORKING SHEETS	☑ YES	☐ NO
PERMITTED ITEM	☐ YES	☑ NO

if yes, items permitted are:

Instructions

Once your exam finishes, you will be given time to scan
a QR code and upload your answers using
your smartphone and laptop.

Here's how to do it.

- This is a **closed book** exam with **Specifically permitted items**. You are not allowed to consult books or online sources during the exam. You may use an unlimited number of blank A4 paper as working sheets.

- A formula sheet is provided within the exam, before the questions.

- There are three types of questions: "short answer questions", "multiple choice question" and "hand-written response".

- Unless otherwise specified, each answer to a "short answer question" on this exam is a rational number; enter your answers as follows. If the answer is an integer then enter it as a numeral. For example 16, 1, or -1 but NOT sixteen, 1.0, minus 1. If the answer is a non-integer rational number then enter it as [numerator]/[denominator] where the fraction is in lowest terms, [numerator] and [denominator] are integers entered as above. For example, -1/2, 5/4 or 11/16 NOT 1/-2, 1.25 or 22/32. Importantly, no answer to a "short answer question" should contain a space.

- The multiple choice question will require you to select one correct answer.

- The hand-written responses will require you to upload your written work using a camera (via a phone or tablet) immediately after the exam is over. The procedure for doing this will be explained to you at the end of the exam.

- Note: Please widen your browser window to see the questions displayed in full size.

- In this assessment, you must **not** use generative artificial intelligence (AI) to generate any materials or content in relation to the assessment task.

- Once the exam duration is finished, your exam will automatically submit. Please ensure you finalise your answers before the end of the allocated exam time.

- **When your exam duration has ended, you are not permitted to write or edit any of your answer sheets.** Please note you must remain under exam conditions and are not permitted to adjust your answer sheets during the upload process once you have submitted your exam.

Instructions

Information

You can review your exam instructions by clicking the 'Show Instructions' button above.

Information

Getting help and support

Assessments without online supervision

If you have any difficulties (technical, medical or otherwise), call the Assessment Support hotline.

If your internet connection drops out during your assessment, don't panic – your responses are automatically saved every 30 seconds.

Assessment Support
Australia: +61 3 9905 4300
Malaysia: +60 3 5514 5600
China: +800 666 274 73

Assessments with online supervision

If you have any difficulties (technical, medical or otherwise) use the Raise Hand button or webchat function to alert your online supervisor.

If you've lost your connection to your online supervisor, call the Assessment Support hotline.

Raise hand

Formula sheet

Information

$a^0 = 1, \quad a^x a^y = a^{x+y}, \quad a^{xy} = (a^x)^y.$

$\log_a(1) = 0, \quad \log_a(xy) = \log_a(x) + \log_a(y), \quad \log_a(x^y) = y\log_a(x), \quad \log_a(x) = \dfrac{\log_b(x)}{\log_b(a)}.$

If $f(x) = x^n$, then $f'(x) = nx^{n-1}$. If $f(x) = a^{cx+d}$, then $f'(x) = c\ln(a)a^{cx+d}$.

If $f(x) = \log_a(cx+d)$, then $f'(x) = \frac{c}{\ln(a)(cx+d)}$.

$$\nabla f(x,y) = \begin{bmatrix} f_x(x,y) \\ f_y(x,y) \end{bmatrix} \qquad H(x,y) = \begin{bmatrix} f_{xx}(x,y) & f_{xy}(x,y) \\ f_{yx}(x,y) & f_{yy}(x,y) \end{bmatrix}.$$

$\det(H(x,y)) > 0$ and $f_{xx}(x,y) > 0 \implies$ local minimum.

$\det(H(x,y)) > 0$ and $f_{xx}(x,y) < 0 \implies$ local maximum.

$\det(H(x,y)) < 0 \implies$ saddle point. $\det(H(x,y)) = 0 \implies$ test inconclusive.

$$\text{If} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{then} \quad \det(A) = ad - bc \quad \text{and} \quad A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Selections of r elements from a set of n elements:

	Unordered	Ordered
With repetition	$\binom{n+r-1}{r}$	n^r
Without repetition	$\dfrac{n!}{r!(n-r)!} = \binom{n}{r}$	$\dfrac{n!}{(n-r)!}$

$$\Pr(A|B) = \frac{\Pr(A\cap B)}{\Pr(B)} \qquad \Pr(A|B) = \frac{\Pr(B|A)\Pr(A)}{\Pr(B|A)\Pr(A) + \Pr(B|\overline{A})\Pr(\overline{A})}.$$

Discrete uniform: $\Pr(X = k) = \frac{1}{b-a+1}$ has $\mathrm{E}[X] = \frac{a+b}{2}$ and $\mathrm{Var}[X] = \frac{(b-a+1)^2-1}{12}$.

Bernoulli: $\Pr(X = k) = \begin{cases} p & \text{for } k = 1 \\ 1-p & \text{for } k = 0 \end{cases}$ has $\mathrm{E}[X] = p$ and $\mathrm{Var}[X] = p(1-p)$.

Geometric: $\Pr(X = k) = p(1-p)^k$ has $\mathrm{E}[X] = \frac{1-p}{p}$ and $\mathrm{Var}[X] = \frac{1-p}{p^2}$.

Binomial: $\Pr(X = k) = \binom{n}{k}p^k(1-p)^{n-k}$ has $\mathrm{E}[X] = np$ and $\mathrm{Var}[X] = np(1-p)$.

Poisson: $\Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ has $\mathrm{E}[X] = \lambda$ and $\mathrm{Var}[X] = \lambda$.

Continuous uniform: $\begin{cases} \frac{1}{b-a} & \text{for } x \in [a,b] \\ 0 & \text{otherwise.} \end{cases}$ has $\mathrm{E}[X] = \frac{a+b}{2}$ and $\mathrm{Var}[X] = \frac{(b-a)^2}{12}$.

Exponential: $\lambda e^{-\lambda x}$ has $\mathrm{E}[X] = \frac{1}{\lambda}$ and $\mathrm{Var}[X] = \frac{1}{\lambda^2}$.

Normal: $\frac{1}{\sqrt{2\sigma^2\pi}}\exp(-\frac{(x-\mu)^2}{2\sigma^2})$ has $\mathrm{E}[X] = \mu$ and $\mathrm{Var}[X] = \sigma^2$.

t-statistic: $t = \frac{\bar{x}-\mu}{s/\sqrt{n}}$

Foundations

Question 1

Evaluate $\sum_{j=2}^{5}(2j+1)$

2 Marks

Question 2

Find the value of the following.

$$\prod_{j=1}^{4} j^2$$

1 Mark

Question 3

Choose which of the following terms correctly applies to the function $f(x) = x^2$

2 Marks

Select one:

- a. Convex
- b. Concave
- c. Both convex and concave
- d. Neither convex nor concave

Question 4

2 Marks

Find the equation of the line through the points $(3, 6)$ and $(6, -3)$. The equation of the line should be of the form $y = mx + b$.

m =

b =

Question 5

2 Marks

Suppose $f(x) = ax^b$, and that a log-log plot (with logarithms taken base 3) of this function shows a straight line with slope 4 and y-intercept 2. What are a and b?

a =

b =

Functions

Question 6

Consider the function $f : [0, \infty) \to \mathbb{R}$ where $f(x) = x^2 + 3x$. What is the value of $f^{-1}(40)$?

2 Marks

Question 7

If $f(x) = e^{-x^2}$, what is $f^{(2)}(0)$?

2 Marks

Question 8

If $f(x) = (x+5)e^{3x}$, find the slope of the tangent line to the graph of f at $x = 0$.

2 Marks

Question 9

3 Marks

Find the maximum value of the product xy, under the constraint that $x + 2y = 80$.

Maximum $=$ ______ at $x =$ ______ and $y =$ ______.

Question 10

For the function $h(x) = e^{2x^2-12x-3}$, find x at which h has a stationary value.

2 Marks

Question 11

Evaluate $\int_{-2}^{4} (\frac{x^3}{4} - 6x) dx$

2 Marks

Question 12

3 Marks

Find the stationary point of the function $f(x, y) = \ln(x^2 + y^2) + \frac{1}{3}x$, with domain $(x, y) \neq (0, 0)$.

The stationary point is $($ ______ , ______ $)$

Question 13

3 Marks

For the function $g(x, y) = (x^2 - y^2)xy$, find the gradient vector ∇g at the point $(2, -1)$.

$\nabla g(2, -1) = ($ ______ , ______ $)$

Question 14

2
Marks

If the gradient of a function f satisfies $\nabla f(x,y) = \begin{pmatrix} 3x^2y^2 - 2x \\ 2x^3y + 6y^2 \end{pmatrix}$, find the determinant of $H(1,-1)$, where H refers to the Hessian matrix.

Question 15

3
Marks

What is the residual sum of squares (RSS) when approximating $(1,7),(2,8),(3,8)$ by $f(x) = 3x + 2$?

RSS =

Linear algebra

Question 16

2 Marks

Let $A = \begin{pmatrix} -1 & 3 \\ -2 & x \end{pmatrix}$. For which $x \in \mathbb{R}$ is A *not* invertible?

Question 17

2 Marks

Let $A = \begin{pmatrix} 0 & -2 & -1 \\ 3 & 1 & 1 \\ 1 & -2 & -1 \end{pmatrix}$. Then $A^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ -4 & -1 & 3 \\ x & 2 & -6 \end{pmatrix}$. Find x.

Question 18

2 Marks

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ -2 & 4 & 2 \\ -1 & -3 & -2 \end{pmatrix}$. One of the eigenvalues of A is 0; if the corresponding eigenvector is of the form $(1, y, z)$, find y and z.

y=

z=

Question 19

2 Marks

Let $a = \begin{pmatrix} -3 \\ 1 \\ -3 \end{pmatrix}$ and $b = \begin{pmatrix} x \\ 3 \\ -3 \end{pmatrix}$. Suppose that a and b are orthogonal. What is x?

Question 20

2 Marks

Let x be the norm of the vector $(1, 3, -7)$.

Find $x^2 =$

Question 21

3

Marks

Solve the following system of linear equations using Gaussian elimination:

$$\begin{pmatrix} -7 & -3 & 3 \\ 2 & 2 & 2 \\ -1 & -4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -16 \\ 0 \\ 12 \end{pmatrix}$$

x=

y=

z=

Combinatorics

Question 22

How many 3 digit numbers are there which are made from 3 different digits? Note the the first digit cannot be 0.

2 Marks

Question 23

A basketball team has 8 people on it. How many different ways are there to form a group of 5 of them to play?

2 Marks

Question 24

There are 27 letters in the Spanish alphabet. If we form a password 8 letters long using only lower case letters from the Spanish alphabet, then there are exactly 3^n such passwords. Find n.

2 Marks

Question 25

A pizza place sells 5 different types of pizza. If we go there to buy 8 pizzas, in how many different ways can we choose these 8 pizzas?

2 Marks

Probability and graphs

Question 26

2 Marks

Random variable X is defined to be the value when a number is chosen at random from 1 to 10, with a uniformly distribution. Y is defined to be 2 when X is even, and 4 when X is odd. Find the expectation $E[2X - Y]$.

Question 27

3 Marks

A game is played in which a fair coin is tossed three times. If 3 heads or 3 tails are thrown, the player wins 40 dollars, while if any other combination occurs, the player loses 4 dollars.

What is the probability of winning? (Write as a decimal to 2 decimal places, not as a fraction)

What is the expected value of their winnings, in dollars? $ (Use a negative number if the expected value of the game is a loss).

Question 28

3 Marks

For the continuous random variable X there is a pdf $h(x)$ for which $h(x) = kx^2(1-x)$ for $0 < x < 1$, and $h(x) = 0$ for x outside this interval. Find k.

Question 29

2 Marks

A graph has a degree sequence of $(1, 2, 3, 3, 3, 3, 3)$. How many edges does it have?

Question 30

4 Marks

The graph G has the adjacency matrix $M = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

How many vertices does G have?

How many edges does G have?

How many components (i.e. separate pieces) does G have?

Question 31

2
Marks

Suppose a random variable X admits the pdf $h(x) = \frac{3}{4}(1 - (x-1)^2)$ for $0 < x < 2$ and 0 elsewhere. Find $E[X]$ and $Var(X)$; remember to reduce all fractions.

$E[X] =$

$Var(X) =$

Long answer questions

Question 32

Explain how to compute the global extrema of $f(x) = x^3 - 6x$ with domain $[-3, 2]$ and find all global extrema.

6
Marks

Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have **extra time** to access your phone to scan a **QR code** and **upload** your answer.
- Clearly label each page with **Student ID** and **this question number** (and **sub part** if applicable) (for example, 'Question 7a')
- **Do not** write your Name on it

No. of answer sheets: 1

Question 33

Let $f(x, y) = 3xy^2 + y^3 - 3x^2 - 3y^2 + 4$. Determine the gradient i.e ∇f, and hence find the Hessian matrix of $f(x, y)$. Find the stationary points, and use the Hessian to classify as much as possible the nature of the stationary points of f (note: if the Hessian test fails just say so, you do not need to try other tests).

6
Marks

Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have **extra time** to access your phone to scan a **QR code** and **upload** your answer.
- Clearly label each page with **Student ID** and **this question number** (and **sub part** if applicable) (for example, 'Question 7a')
- **Do not** write your Name on it

No. of answer sheets: 1

Question 34

Consider the following matrix.

6
Marks

$$A = \begin{bmatrix} 2 & -2 \\ -3 & 1 \end{bmatrix}$$

a) Find the eigenvalues of A.

b) Find eigenvectors corresponding to each of the eigenvalues.

c) Explain how you would find matrices P, D with P invertible and D diagonal so that $A = PDP^{-1}$. You should give explicit values for P and D, but you do not need to calculate P^{-1}.

Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have **extra time** to access your phone to scan a **QR code** and **upload** your answer.
- Clearly label each page with **Student ID** and **this question number** (and **sub part** if applicable) (for example, 'Question 7a')
- **Do not** write your Name on it

No. of answer sheets: 1

Question 35

6

Marks

Consider the following system of linear equations in matrix form:

$$\begin{bmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 2 & 1 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ u \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$$

Solve the system and write down all solutions (x, y, z, u)

Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have **extra time** to access your phone to scan a **QR code** and **upload** your answer.
- Clearly label each page with **Student ID** and **this question number** (and **sub part** if applicable) (for example, 'Question 7a')
- **Do not** write your Name on it

No. of answer sheets: 1

Question 36

6

Marks

The weather forecaster has predicted rain for tomorrow. When it actually rains, the forecaster had correctly forecasted rain 80% of the time; however, when it doesn't rain they had incorrectly forecasted rain 10% of the time. It rains 20% of the time. What is the probability that it will rain tomorrow?

Please answer the question on your blank piece of paper.

- After your exam finishes, you'll have **extra time** to access your phone to scan a **QR code** and **upload** your answer.
- Clearly label each page with **Student ID** and **this question number** (and **sub part** if applicable) (for example, 'Question 7a')
- **Do not** write your Name on it

No. of answer sheets: 1